

Data Privacy: Homeworks #4 Solutions

1. **Neyman–Pearson Lemma.** Let T be a real-valued random variable. Under hypothesis H_b (for $b \in \{0, 1\}$), assume T has a continuous density p_b on $\mathbb{R}$, with $p_0(t) > 0$ for every t . Define the *likelihood ratio*

$$\Lambda(t) = \frac{p_1(t)}{p_0(t)}.$$

A *test* is a function $\phi : \mathbb{R} \rightarrow [0, 1]$, where $\phi(t)$ is the probability of rejecting H_0 after observing $T = t$. Define

$$\alpha(\phi) = \mathbb{E}_{H_0}[\phi(T)] \quad (\text{size, type-I error}), \quad \beta(\phi) = \mathbb{E}_{H_1}[\phi(T)] \quad (\text{power}).$$

Fix a level $\alpha_0 \in (0, 1)$, and *assume that the function* $G(\tau) := \Pr_{H_0}[\Lambda(T) > \tau]$ *is continuous on* $[0, \infty)$. Prove the following three claims.

- (a) **Existence.** Show that there exists a threshold $\tau^* \geq 0$ such that the test

$$\phi^*(t) = \mathbf{1}[\Lambda(t) > \tau^*]$$

satisfies $\alpha(\phi^*) = \alpha_0$ exactly.

Hint: Show that G is non-increasing on $[0, \infty)$ with $G(0) \leq 1$ and $G(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$. Then apply the intermediate value theorem.

- (b) **Optimality.** Show that for every test ϕ with $\alpha(\phi) \leq \alpha_0$, we have $\beta(\phi) \leq \beta(\phi^*)$.

Hint: Verify that for every t , $(\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) \geq 0$, then integrate over $\mathbb{R}$.

- (c) **Uniqueness.** Suppose ϕ is any most-powerful level- α_0 test (i.e., $\alpha(\phi) \leq \alpha_0$ and $\beta(\phi) = \beta(\phi^*)$). Show that $\phi(t) = \phi^*(t)$ for every t with $\Lambda(t) \neq \tau^*$, except possibly on a set of probability 0.

Hint: The integrand from part (b) is non-negative and now integrates to 0.

Solution:

- (a) **Existence of τ^* .** If $\tau_1 \leq \tau_2$ then $\{\Lambda > \tau_2\} \subseteq \{\Lambda > \tau_1\}$, so $G(\tau_2) \leq G(\tau_1)$. Hence G is non-increasing on $[0, \infty)$. Also,

$$G(0) = \Pr_{H_0}[\Lambda(T) > 0] \leq 1, \quad G(\tau) \xrightarrow{\tau \rightarrow \infty} 0,$$

because $\{\Lambda > \tau\}$ shrinks to the empty event as $\tau \rightarrow \infty$. By the assumed continuity of G and the intermediate value theorem, there exists $\tau^* \geq 0$ with $G(\tau^*) = \alpha_0$. Then

$$\alpha(\phi^*) = \mathbb{E}_{H_0}[\mathbf{1}[\Lambda(T) > \tau^*]] = \Pr_{H_0}[\Lambda(T) > \tau^*] = G(\tau^*) = \alpha_0.$$

- (b) **Optimality.** For each fixed t we check the pointwise inequality $(\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) \geq 0$ case by case:

- If $p_1(t) > \tau^* p_0(t)$: $\phi^*(t) = 1 \geq \phi(t)$ and the second factor is > 0 , so the product is ≥ 0 .
- If $p_1(t) < \tau^* p_0(t)$: $\phi^*(t) = 0 \leq \phi(t)$ and the second factor is < 0 , so the product is ≥ 0 .
- If $p_1(t) = \tau^* p_0(t)$: the second factor is 0.

Integrate the inequality over $\mathbb{R}$:

$$0 \leq \int (\phi^*(t) - \phi(t))(p_1(t) - \tau^* p_0(t)) dt = (\beta(\phi^*) - \beta(\phi)) - \tau^*(\alpha(\phi^*) - \alpha(\phi)).$$

Since $\alpha(\phi^*) = \alpha_0 \geq \alpha(\phi)$ and $\tau^* \geq 0$, the second term on the right is non-negative, so $\beta(\phi^*) - \beta(\phi) \geq \tau^*(\alpha_0 - \alpha(\phi)) \geq 0$.

- (c) **Uniqueness.** If ϕ is also most powerful, $\beta(\phi) = \beta(\phi^*)$, and combining with $\alpha(\phi) \leq \alpha_0 = \alpha(\phi^*)$,

$$0 \leq \int (\phi^* - \phi)(p_1 - \tau^* p_0) dt = -\tau^*(\alpha_0 - \alpha(\phi)) \leq 0,$$

so the integral equals 0. The integrand is ≥ 0 everywhere, and its integral is 0, so it must equal 0 except possibly on a set of probability 0. On $\{t : p_1(t) > \tau^* p_0(t)\}$ the second factor is strictly positive, forcing $\phi^*(t) - \phi(t) = 0$ there (except on a probability-zero set); since $\phi^* = 1$ on this set, $\phi = 1$ there. Symmetrically, $\phi = 0 = \phi^*$ on $\{p_1 < \tau^* p_0\}$. Hence ϕ agrees with ϕ^* on $\{\Lambda \neq \tau^*\}$ (apart from a probability-zero set).

2. **DP $\Rightarrow$ MIA Bounds.** Let $M : \mathcal{X}^n \rightarrow \mathcal{Y}$ be (ε, δ) -differentially private, and let D_0, D_1 be two neighboring datasets. Consider any (possibly randomized) binary test $\mathcal{A} : \mathcal{Y} \rightarrow \{0, 1\}$ that tries to distinguish $M(D_0)$ from $M(D_1)$. Write

$$\text{FPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_0)) = 1], \quad \text{TPR}(\mathcal{A}) = \Pr[\mathcal{A}(M(D_1)) = 1].$$

- (a) Show that

$$\text{TPR}(\mathcal{A}) \leq e^\varepsilon \text{FPR}(\mathcal{A}) + \delta, \quad 1 - \text{FPR}(\mathcal{A}) \leq e^\varepsilon (1 - \text{TPR}(\mathcal{A})) + \delta.$$

- (b) Define the MI advantage $\text{Adv}(\mathcal{A}) = \text{TPR}(\mathcal{A}) - \text{FPR}(\mathcal{A})$. Show the loose bound

$$\text{Adv}(\mathcal{A}) \leq e^\varepsilon - 1 + \delta.$$

Then, by combining both inequalities of part (a), derive the tighter Kairouz–Oh–Viswanath bound

$$\text{Adv}(\mathcal{A}) \leq \frac{e^\varepsilon - 1 + 2\delta}{e^\varepsilon + 1}.$$

- (c) Conclude that for $\varepsilon = 0.1$ and $\delta = 10^{-5}$, no adversary can exceed advantage ≈ 0.105 , but for $\varepsilon = 8$ the bound is vacuous.

Solution:

- (a) Let $S = \{y : \mathcal{A}(y) = 1\} \subseteq \mathcal{Y}$ (if $\mathcal{A}$ is randomized, average over its internal randomness; this does not affect the DP inequality). Apply the (ε, δ) -DP guarantee to S with the ordered pair (D_1, D_0) :

$$\Pr[M(D_1) \in S] \leq e^\varepsilon \Pr[M(D_0) \in S] + \delta \iff \text{TPR} \leq e^\varepsilon \text{FPR} + \delta.$$

Apply it to the complementary event S^c with the ordered pair (D_0, D_1) :

$$\Pr[M(D_0) \in S^c] \leq e^\varepsilon \Pr[M(D_1) \in S^c] + \delta \iff 1 - \text{FPR} \leq e^\varepsilon (1 - \text{TPR}) + \delta.$$

- (b) **Loose bound.** From the first inequality and $\text{FPR} \leq 1$,

$$\text{Adv} = \text{TPR} - \text{FPR} \leq (e^\varepsilon - 1)\text{FPR} + \delta \leq e^\varepsilon - 1 + \delta.$$

Tighter (Kairouz–Oh–Viswanath). The pair of inequalities from (a) defines a polygonal region in the (FPR, TPR) -plane. The maximum of $\text{TPR} - \text{FPR}$ over this region is attained where both inequalities are tight. Rewriting the second inequality as

$$\text{TPR} \geq 1 - e^{-\varepsilon}(1 - \text{FPR}) - \delta e^{-\varepsilon},$$

and setting equal to $e^\varepsilon \text{FPR} + \delta$, we get

$$(e^\varepsilon - e^{-\varepsilon})\text{FPR} = (1 - e^{-\varepsilon})(1 - \delta) \implies \text{FPR}^* = \frac{1 - \delta}{e^\varepsilon + 1}.$$

Then $\text{TPR}^* = e^\varepsilon \text{FPR}^* + \delta = (e^\varepsilon + \delta)/(e^\varepsilon + 1)$, so

$$\text{Adv}^* = \text{TPR}^* - \text{FPR}^* = \frac{e^\varepsilon + \delta - (1 - \delta)}{e^\varepsilon + 1} = \frac{e^\varepsilon - 1 + 2\delta}{e^\varepsilon + 1}.$$

- (c) For $\varepsilon = 0.1, \delta = 10^{-5}$: loose bound $= e^{0.1} - 1 + 10^{-5} \approx 0.1052$; tighter bound ≈ 0.0500 . For $\varepsilon = 8$: $e^8 \approx 2981$, so the loose bound is ≈ 2980 and the tighter bound is ≈ 1 . Both are useless — advantage is automatically ≤ 1 .

3. **Generalization Gap as a Leakage Lower Bound (Yeom).** Suppose the loss is bounded: $\ell(f_\theta, z) \in [0, B]$ for all θ, z . Define

$$R_{\text{train}} = \mathbb{E}_{z \sim \text{Unif}(D)}[\ell(f_\theta, z)], \quad R_{\text{pop}} = \mathbb{E}_{z \sim \mathcal{P}}[\ell(f_\theta, z)], \quad \Delta = R_{\text{pop}} - R_{\text{train}} \geq 0.$$

The MIA game: sample $b \sim \text{Unif}\{0, 1\}$. If $b = 1$, return z uniform over D ; otherwise, return $z \sim \mathcal{P}$. The adversary uses a threshold attack $\mathcal{A}_\tau(z) = \mathbf{1}[\ell(f_\theta, z) \leq \tau]$. Let $F_1(\tau) = \Pr[\ell \leq \tau \mid b = 1]$ and $F_0(\tau) = \Pr[\ell \leq \tau \mid b = 0]$; assume both have continuous densities on $[0, B]$.

- (a) Show that the advantage of $\mathcal{A}_\tau$ is

$$\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau).$$

- (b) Prove the *layer-cake identity*: for any random variable X with continuous density on $[0, B]$,

$$\mathbb{E}[X] = \int_0^B (1 - F_X(t)) dt.$$

Then deduce

$$\int_0^B (F_1(t) - F_0(t)) dt = \Delta.$$

- (c) Conclude that there exists a threshold $\tau^* \in [0, B]$ for which the threshold attack achieves advantage

$$\text{Adv}(\mathcal{A}_{\tau^*}) \geq \frac{\Delta}{B}.$$

In words: the generalization gap is a *lower* bound on the leakage of the best threshold attack — not an upper bound, contrary to a common informal slogan.

Hint: $\Delta = \int_0^B (F_1 - F_0) dt \leq B \cdot \sup_t (F_1(t) - F_0(t))$.

- (d) Now suppose additionally that under each hypothesis the loss is Gaussian:

$$\ell \mid b = 1 \sim \mathcal{N}(R_{\text{train}}, \sigma^2), \quad \ell \mid b = 0 \sim \mathcal{N}(R_{\text{pop}}, \sigma^2).$$

(Ignore the issue that Gaussians are not literally bounded; the formula below is the correct one for the full range $\mathbb{R}$.) Show that the maximum-advantage threshold attack has $\tau^* = (R_{\text{train}} + R_{\text{pop}})/2$ and achieves

$$\text{Adv}^* = 2\Phi\left(\frac{\Delta}{2\sigma}\right) - 1.$$

For small Δ/σ , derive the linear approximation $\text{Adv}^* \approx \Delta/(\sigma\sqrt{2\pi})$ and explain why noisier loss distributions reduce leakage.

Solution:

- (a) By definitions, $\text{TPR} = \Pr[\mathcal{A}_\tau = 1 \mid b = 1] = \Pr[\ell \leq \tau \mid b = 1] = F_1(\tau)$ and $\text{FPR} = F_0(\tau)$. Subtracting gives $\text{Adv}(\mathcal{A}_\tau) = F_1(\tau) - F_0(\tau)$.
- (b) **Layer cake.** Let f_X be the density of X . Write $t = \int_0^t 1 ds$ and substitute:

$$\mathbb{E}[X] = \int_0^B t f_X(t) dt = \int_0^B \left(\int_0^t 1 ds \right) f_X(t) dt = \int \int_{0 \leq s < t \leq B} f_X(t) ds dt.$$

Switch the order of integration over the triangle $\{0 \leq s < t \leq B\} = \{0 \leq s \leq B, s < t \leq B\}$:

$$\mathbb{E}[X] = \int_0^B \left(\int_s^B f_X(t) dt \right) ds = \int_0^B \Pr[X > s] ds = \int_0^B (1 - F_X(s)) ds.$$

Apply this to ℓ under each hypothesis:

$$R_{\text{pop}} - R_{\text{train}} = \int_0^B [(1 - F_0(t)) - (1 - F_1(t))] dt = \int_0^B (F_1(t) - F_0(t)) dt = \Delta.$$

- (c) Let $h(t) = F_1(t) - F_0(t)$. If $h(t) < \Delta/B$ for every $t \in [0, B]$, then $\int_0^B h dt < B \cdot (\Delta/B) = \Delta$, contradicting (b). Hence some $\tau^* \in [0, B]$ satisfies $h(\tau^*) \geq \Delta/B$, so $\text{Adv}(\mathcal{A}_{\tau^*}) \geq \Delta/B$. The generalization gap is therefore a *lower* bound on the leakage of the best threshold attack, not an upper bound.
- (d) **Gaussian.** With $\mu_1 = R_{\text{train}}, \mu_0 = R_{\text{pop}}$ we have $F_b(t) = \Phi((t - \mu_b)/\sigma)$, so

$$h(\tau) = \Phi\left(\frac{\tau - R_{\text{train}}}{\sigma}\right) - \Phi\left(\frac{\tau - R_{\text{pop}}}{\sigma}\right).$$

Setting $h'(\tau) = 0$ gives $\phi((\tau - R_{\text{train}})/\sigma) = \phi((\tau - R_{\text{pop}})/\sigma)$ (where ϕ is the standard normal pdf), so $|\tau - R_{\text{train}}| = |\tau - R_{\text{pop}}|$, i.e., $\tau^* = (R_{\text{train}} + R_{\text{pop}})/2$. Then $\tau^* - R_{\text{train}} = \Delta/2$ and $\tau^* - R_{\text{pop}} = -\Delta/2$, so

$$\text{Adv}^* = \Phi(\Delta/(2\sigma)) - \Phi(-\Delta/(2\sigma)) = 2\Phi(\Delta/(2\sigma)) - 1.$$

For small $x = \Delta/(2\sigma)$, $\Phi(x) \approx 1/2 + x\phi(0) = 1/2 + x/\sqrt{2\pi}$, so $\text{Adv}^* \approx 2x/\sqrt{2\pi} = \Delta/(\sigma\sqrt{2\pi})$. Larger σ (noisier loss distributions) flattens the gap and reduces leakage — the same idea behind DP noise injection.

4. **Influence-Function Unlearning (Closed Form).** Let $L(\theta; D) = \frac{1}{n} \sum_{i=1}^n \ell(\theta; z_i)$ be a twice-differentiable, λ -strongly convex loss in $\theta \in \mathbb{R}^d$. Recall that λ -strong convexity means the Hessian $H_\theta = \nabla^2 L(\theta; D)$ satisfies $H_\theta \succeq \lambda I$, which in turn implies H_θ is invertible with $\|H_\theta^{-1}\|_{\text{op}} \leq 1/\lambda$. Let $\theta_o = \arg \min_\theta L(\theta; D)$ and $\theta_{-z} = \arg \min_\theta L(\theta; D \setminus \{z\})$.

- (a) Consider the parametric family

$$L_t(\theta) = L(\theta; D) - \frac{t}{n} \ell(\theta; z), \quad t \in [0, 1],$$

and let $\theta(t) = \arg \min_\theta L_t(\theta)$. (Note that $L_0 = L(\cdot; D)$ gives $\theta(0) = \theta_o$, and L_1 is a positive scalar multiple of $L(\cdot; D \setminus \{z\})$, so $\theta(1) = \theta_{-z}$.) Use the first-order optimality condition $\nabla_\theta L_t(\theta(t)) = 0$ and differentiate with respect to t (the *implicit function theorem*) to derive

$$\left. \frac{d\theta(t)}{dt} \right|_{t=0} = \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

- (b) Conclude the first-order Newton-step approximation

$$\theta_{-z} - \theta_o \approx \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z).$$

The “unlearned” parameter is then $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_\theta \ell(\theta_o; z)$.

- (c) Assume in addition that ℓ is G -Lipschitz in θ (so $\|\nabla_\theta \ell(\theta; z)\| \leq G$ for all θ, z). Show that the L_2 -sensitivity of the unlearned parameter to the choice of deleted point is bounded:

$$\sup_{z, z'} \|\theta_u(D, z) - \theta_u(D, z')\| \leq \frac{2G}{\lambda n}.$$

(Treat $H_{\theta_o}^{-1}$ as fixed; its operator norm is $\leq 1/\lambda$ by strong convexity. This is the first-order sensitivity used in certified unlearning.)

Solution:

- (a) **Implicit derivative.** By strong convexity the unique minimizer $\theta(t) = \arg \min L_t$ exists. The first-order condition is

$$F(\theta(t), t) := \nabla_{\theta} L(\theta(t); D) - \frac{t}{n} \nabla_{\theta} \ell(\theta(t); z) = 0.$$

Differentiate both sides with respect to t (chain rule):

$$\left[\nabla^2 L(\theta(t); D) - \frac{t}{n} \nabla^2 \ell(\theta(t); z) \right] \theta'(t) - \frac{1}{n} \nabla_{\theta} \ell(\theta(t); z) = 0.$$

At $t = 0$, the bracket equals $\nabla^2 L(\theta_o; D) = H_{\theta_o}$, and $\theta(0) = \theta_o$, so

$$\theta'(0) = \frac{1}{n} H_{\theta_o}^{-1} \nabla_{\theta} \ell(\theta_o; z).$$

(Invertibility of H_{θ_o} follows from $H_{\theta_o} \succeq \lambda I$.)

- (b) **Taylor approximation.** A first-order Taylor expansion of $\theta(t)$ around $t = 0$ at $t = 1$ gives

$$\theta_{-z} - \theta_o = \theta(1) - \theta(0) \approx \theta'(0) \cdot 1 = \frac{1}{n} H_{\theta_o}^{-1} \nabla_{\theta} \ell(\theta_o; z),$$

so $\theta_u(D, z) = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla_{\theta} \ell(\theta_o; z)$ is the Newton-step formula from the slides.

- (c) **Sensitivity.** For two candidate deleted points z, z' :

$$\theta_u(D, z) - \theta_u(D, z') = \frac{1}{n} H_{\theta_o}^{-1} (\nabla_{\theta} \ell(\theta_o; z) - \nabla_{\theta} \ell(\theta_o; z')).$$

Take norms, using $\|H_{\theta_o}^{-1}\|_{\text{op}} \leq 1/\lambda$ and $\|\nabla_{\theta} \ell\| \leq G$:

$$\|\theta_u(D, z) - \theta_u(D, z')\| \leq \frac{1}{\lambda n} (G + G) = \frac{2G}{\lambda n}.$$

This is the L_2 -sensitivity used to calibrate Gaussian noise in certified unlearning.

5. **Certified (ε, δ) -Unlearning by Gaussian Noise.** Building on the previous problem, let $A(D) = \arg \min_{\theta} L(\theta; D)$ be the (deterministic) full-retrain map, and define the noisy first-order unlearning algorithm

$$\mathcal{U}(D, z) = \theta_o(D) + \frac{1}{n} H_{\theta_o(D)}^{-1} \nabla_{\theta} \ell(\theta_o(D); z) + N, \quad N \sim \mathcal{N}(0, \sigma^2 I_d).$$

Assume ℓ is λ -strongly convex and G -Lipschitz, and that the first-order approximation in Problem 7(b) has L_2 error bounded by

$$\eta := \|\theta_u^{1\text{st}} - A(D \setminus \{z\})\| \leq \frac{C_2 G^2}{\lambda^2 n^2}$$

for some universal constant C_2 (a known second-order bound that you may use without proof).

- (a) Show that the deterministic part of $\mathcal{U}(D, z)$ differs from $A(D \setminus \{z\})$ in L_2 norm by at most

$$\eta \leq \frac{C_2 G^2}{\lambda^2 n^2}.$$

- (b) Recall the Gaussian-mechanism privacy bound: if $f, g : \mathcal{X}^n \rightarrow \mathbb{R}^d$ satisfy $\|f(X) - g(X)\| \leq \eta$ for every X , then $f(X) + N$ and $g(X) + N$ are (ε, δ) -indistinguishable provided

$$\sigma \geq \frac{\eta}{\varepsilon} \sqrt{2 \log(1.25/\delta)}.$$

Apply this with $f = \theta_u^{1\text{st}}$ and $g = A(\cdot \setminus \{z\})$ to conclude that

$$\sigma \geq \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon} \sqrt{2 \log(1.25/\delta)}$$

yields an (ε, δ) -certified unlearning algorithm.

- (c) How does the required noise scale with n ? Compare to the Gaussian-mechanism noise needed for (ε, δ) -DP of the trained model itself ($\sigma \propto 1/(n\varepsilon)$). Explain in one or two sentences why the unlearning noise is smaller for large n .

Solution:

- (a) **Sensitivity.** Define $\theta_u^{1\text{st}} = \theta_o + \frac{1}{n} H_{\theta_o}^{-1} \nabla \ell(\theta_o; z)$, so $\mathcal{U}(D, z) = \theta_u^{1\text{st}} + N$. The exact retrain is $\theta_{-z} = A(D \setminus \{z\})$. The hypothesis says

$$\|\theta_u^{1\text{st}} - \theta_{-z}\| \leq \eta \leq \frac{C_2 G^2}{\lambda^2 n^2}.$$

So the deterministic part of $\mathcal{U}$ differs from $A(D \setminus \{z\})$ by at most η in L_2 .

- (b) **Gaussian-mechanism bound.** Apply the cited Gaussian-mechanism guarantee with $f = \theta_u^{1\text{st}}$ and $g = \theta_{-z}$, both viewed as functions of the input (D, z) :

$$\sigma \geq \frac{\eta}{\varepsilon} \sqrt{2 \log(1.25/\delta)} \iff \sigma \geq \frac{C_2 G^2}{\lambda^2 n^2 \varepsilon} \sqrt{2 \log(1.25/\delta)}.$$

This matches the Guo et al. (2020) bound.

- (c) Unlearning noise scales as $\sigma \propto 1/n^2$, whereas DP-of-the-trained-model noise scales as $\sigma_{\text{DP}} \propto 1/n$. The unlearning task is *easier* for large n : a single training point's influence on the optimum is itself $O(1/n)$, and the Newton correction in Problem 7 reduces the residual to $O(1/n^2)$ — much smaller than what full DP would need to mask.